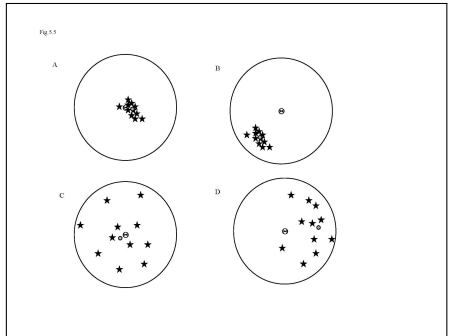
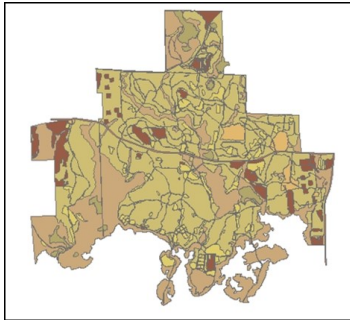


Sampling and Estimation



Key concepts

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- Parameters are almost never known. Why?
 - ▶ We usually have to sample
 - ▶ Animals are hard to detect
- Good sampling designs yield accurate estimates of unknown parameters.

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A characteristic of a dataset (often used to estimate a parameter).

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	Population parameter	Parameter estimate
Population size	N_t	$\hat{N}_t$
Growth rate	r	$\hat{r}$
Occurrence probability	ψ	$\hat{\psi}$

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Target population

The population of interest. Should be defined in terms of time and space.

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Sampled population

The sampled portion of the population of interest, usually defined in terms of the sample units (such as plots, quadrats, etc.).

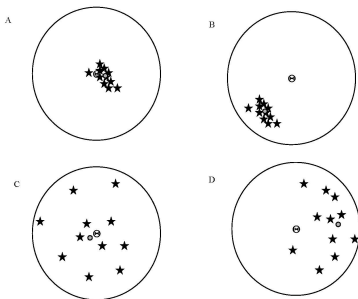
Accuracy has two components:

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Fig 5.5



HOW DO WE REDUCE VARIANCE?

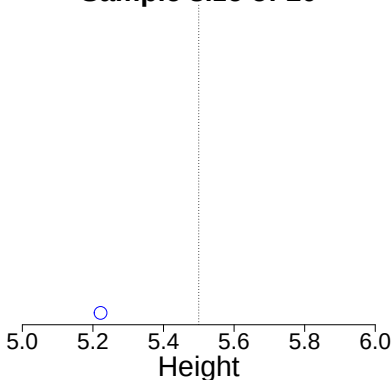
Huge sample size

EFFECT OF SAMPLE SIZE

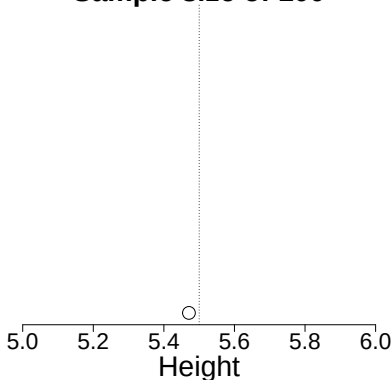
Suppose we want to estimate the height of students on campus, and we have enough resources to repeat a survey many times.

Each point below is an estimate.

Sample size of 10



Sample size of 100

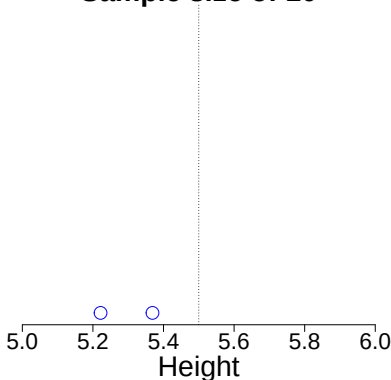


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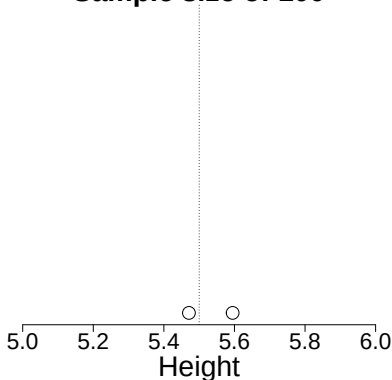
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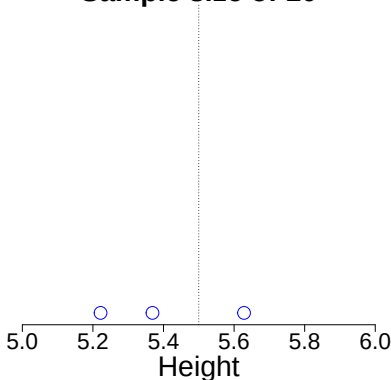


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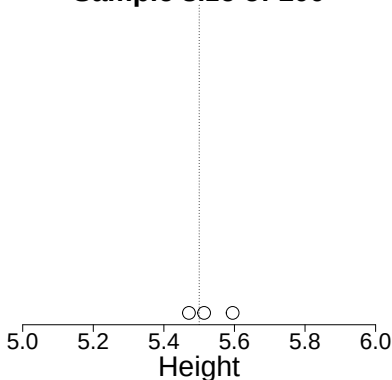
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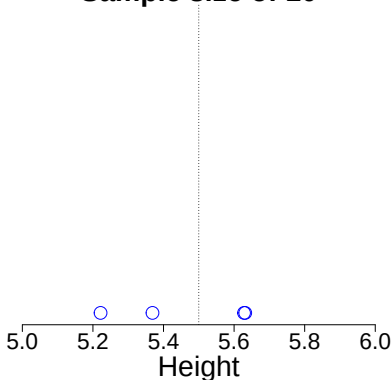


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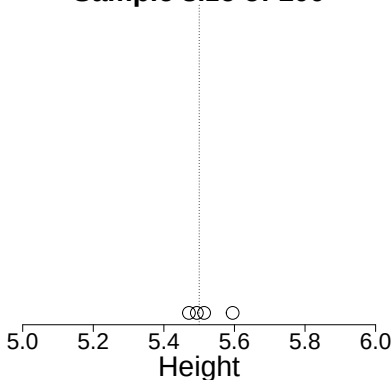
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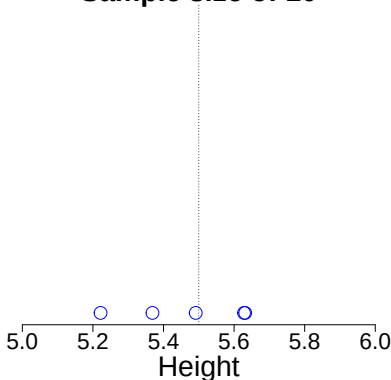


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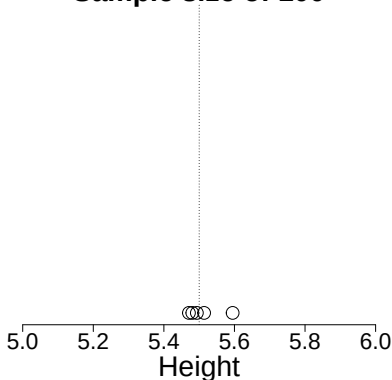
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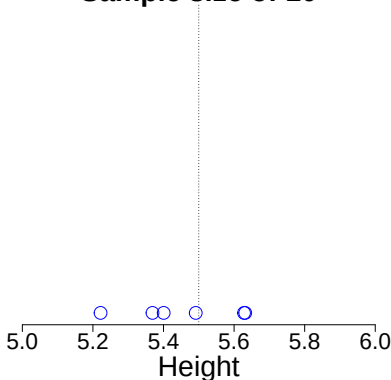


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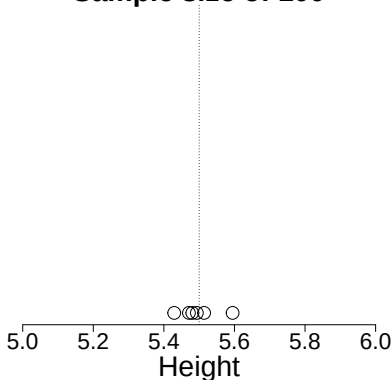
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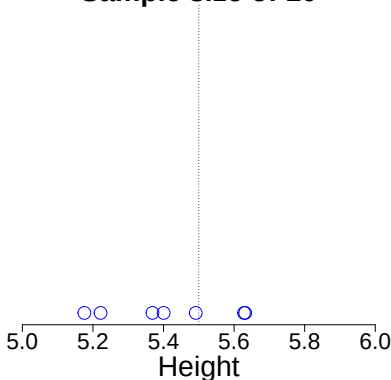


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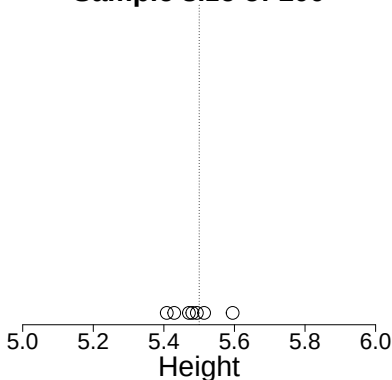
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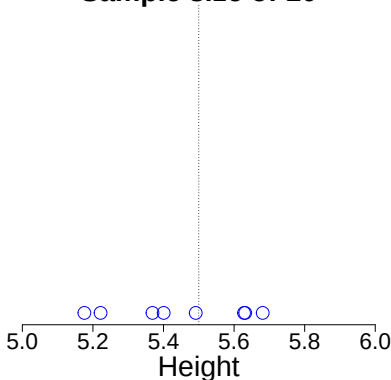


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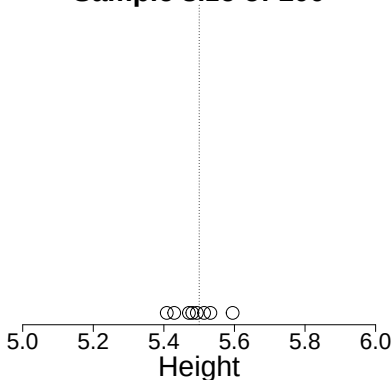
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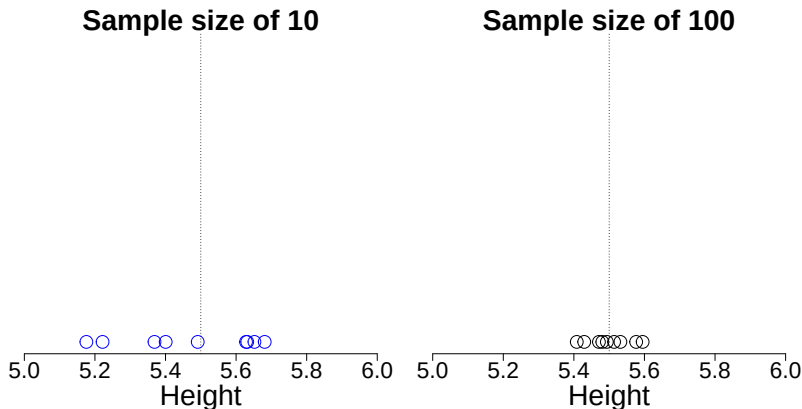
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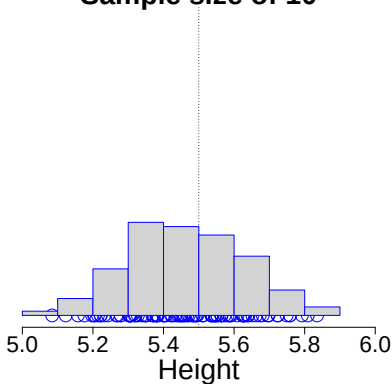


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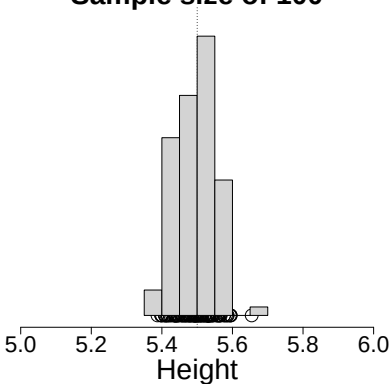
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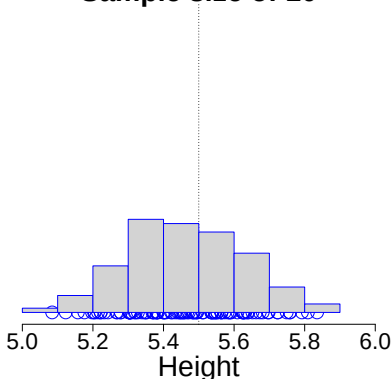


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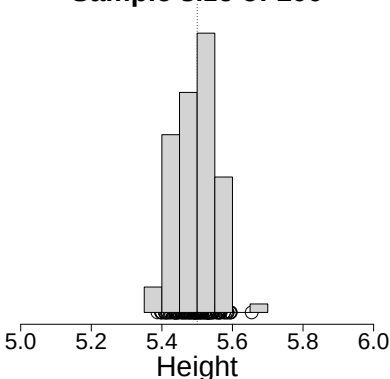
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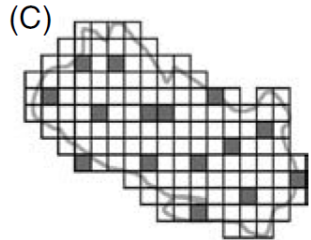
The standard deviation of the sampling distribution is called the standard error (SE)

HOW DO WE REDUCE BIAS?

Randomization

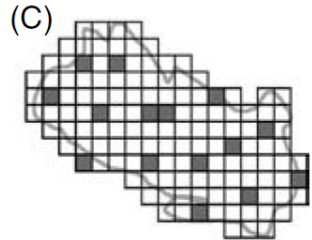
Simple random sampling

- All sample units have the same inclusion probability



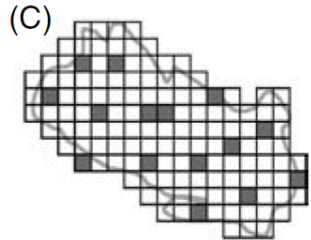
Simple random sampling

- All sample units have the same inclusion probability
- Easiest and most reliable method



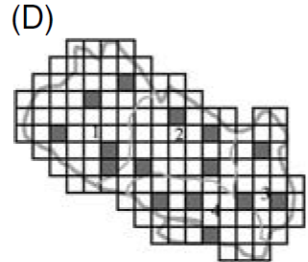
Simple random sampling

- All sample units have the same inclusion probability
- Easiest and most reliable method
- But not always cost effective



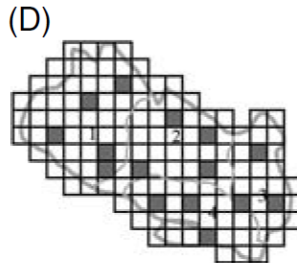
Stratified random sampling

- Useful when study area is characterized by several homogeneous regions



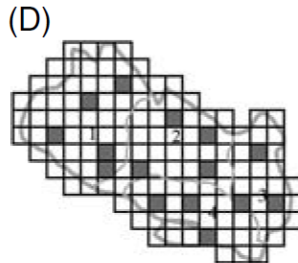
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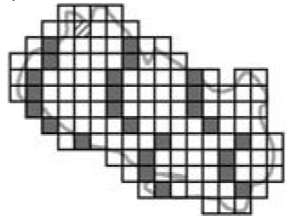
- Useful when study area is characterized by several homogeneous regions
- Regions with higher variability should be sampled more intensively than regions with low variability
- Often more cost effective than simple random sampling



Systematic sampling

- Sample units are selected according to a regular, ordered scheme with the first unit being sampled randomly.

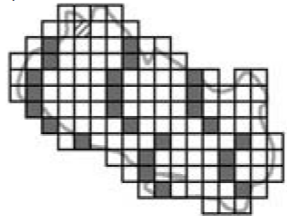
(E)



Systematic sampling

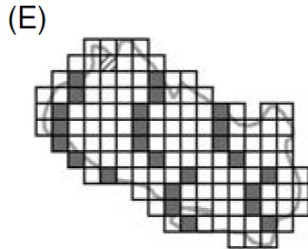
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(E)



Systematic sampling

- Sample units are selected according to a regular, ordered scheme with the first unit being sampled randomly.
- Easy to implement in the field
- Potentially dangerous because sample unit spacing could coincide with natural spacing of environmental features



Main points

- We have to estimate model parameters
- Reliable estimates require good sampling design
- Replication reduces variance
- Randomization reduces bias